College of Artificial Intelligence (El Alamein)

LM BOX 5 – CLASSIC GAMES POWERED BY COMPUTER VISION

Computer Vision (IN412): Final Project Proposal

Names: Mohamed Youssef – Abdelrhman Saeed

Registration Numbers: 211001821 - 211003941

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1. Project Description:

- LM Box 5 combines retro gaming with computer vision, allowing two players to compete in 3 classic games—ping pong, fruit ninja, and a runner game. The project includes a UI for game selection, settings, and face capture to enhance user interaction.
- The project integrates traditional gaming with modern computer vision, highlighting how vision-based inputs can replace controllers. This fusion demonstrates the potential of computer vision in creating interactive, immersive experiences.
- The project tackles real-time object detection, motion tracking, and face detection, primarily using YOLO models. Each game poses unique challenges, making it a valuable testbed for computer vision applications in dynamic environments.

2. Problem Statement:

- LM Box 5 addresses the challenge of integrating real-time computer vision into interactive, two-player games. Traditional games rely on controllers, but this project uses vision-based detection to recognize and track players' movements.
- The projects align with the course's focus on applying computer vision techniques in practical scenarios by providing a hands-on way to learn about model performance and adaptability in fast-paced environments, deepening understanding of vision applications in dynamic contexts.

3. Methodology:

- LM Box 5 will use Python, Jupyter Notebook, and libraries like NumPy, OpenCV, PyGame, and Ultralytics. These tools will support tasks such as image processing, object detection, and game development.
- OpenCV and Ultralytics' YOLO models are ideal for real-time object detection, which is crucial for tracking players' movements. PyGame facilitates game creation and interaction, while Jupyter Notebook provides a flexible environment for developing and refining computer vision models. This setup is well-suited for a project focused on responsive, vision-based gaming applications.

4. Data:

- LM Box 5 will utilize three publicly available YOLO datasets: pose estimation (COCO or MPII Human Pose), hand joint detection (HandKeypoints or Oxford Hand Dataset), and hand gesture detection (Gesture Recognition Toolkit). These datasets will help train each model for detecting player actions and interactions accurately.

- Pose Estimation Dataset:
  - Source: COCO or MPII Human Pose
  - Size: 25,000 to 40,000 annotated images.
  - Content: Contains annotations for body key points across various poses, capturing diverse human activities.
- Hand Joint Detection Dataset:
  - Source: HandKeypoints or Oxford Hand Dataset
  - Size: 4,000 to 10,000 images.
  - Content: Features annotated key points for finger joints and palm landmarks across various hand poses.
- Hand Gesture Detection Dataset:
  - Source: Gesture Recognition Toolkit or RWTH-PHOENIX-Weather 2014T
  - Size: 2,000 to 5,000 images or video clips.
  - Content: Includes labeled examples of various hand gestures in different conditions.

5. Evaluation Metrics:

- The success of the LM Box 5 computer vision models will be assessed through evaluation metrics that are specific to each of the model's nature. Additionally, performance will be measured based on user feedback to gauge the overall experience and satisfaction.

- Mean Average Precision (mAP): Measures the accuracy of the model in detecting objects across multiple classes, calculated as the average precision across different Intersection over Union thresholds.
- Intersection over Union (IoU): Assesses the overlap between the predicted bounding boxes and the ground truth boxes, providing insights into localization accuracy.
- Frame Per Second (FPS): Measures the model's inference speed, indicating how well it performs in real-time scenarios, which is crucial for interactive gaming.

6. Expected Results:

- By the end of the LM Box 5 project, we aim to create a fully functional interactive gaming platform that accurately detects player movements using computer vision. We expect high mAP scores, and real-time performance suitable for seamless gameplay. Positive user feedback should highlight the effectiveness of vision-based controls.

- Achieving these results will demonstrate the practical application of computer vision in enhancing traditional gaming. The success of LM Box 5 will validate the use of YOLO models for real-time gaming experiences, addressing the challenges outlined in the problem statement.

7. References:

- https://chatgpt.com/
- https://www.pygame.org/docs/
- https://ai.google.dev/edge/mediapipe/solutions/vision/gesture_recognizer/python?authuser=1
- https://www.ultralytics.com/hub
- https://docs.ultralytics.com/datasets/pose/hand-keypoints/
- https://www.kaggle.com/datasets/awsaf49/coco-2017-dataset